To what extent are data centers concentrated in water-stressed regions in the U.S.?

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Abstract

The rapid expansion of Artificial Intelligence (AI) technologies has intensified the environmental footprint of data centers, particularly regarding water consumption for cooling. This study analyzes the geographic overlap between AI data center prevalence and regional water stress across the United States. Using data from the World Resources Institute's Aqueduct Water Risk Atlas and publicly available data center counts, the analysis identifies states where infrastructure development intersects with severe water scarcity. Visualizations reveal that regions such as Arizona, California, and Texas face the highest composite risk, combining high concentrations of AI infrastructure with chronic water stress. These insights provide a data-driven foundation for sustainable infrastructure planning, corporate ESG strategies, and resource management policies.

Introduction

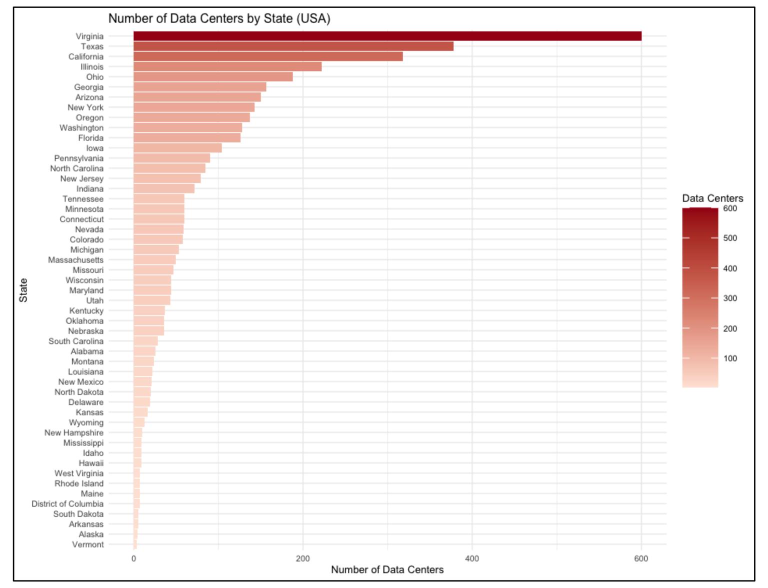
Artificial Intelligence (AI) data centers are proliferating rapidly to support growing computational demands, particularly with the expansion of generative AI applications. These facilities are highly resource-intensive, consuming large amounts of energy and water for cooling purposes. Despite this, data centers are often sited in regions already facing significant water stress, exacerbating local resource scarcity (And The West 2025; Department of Energy 2025).

The most popular U.S. data center locations are concentrated in Northern Virginia and Northern California, with substantial infrastructure in Illinois, New York/New Jersey, and Texas (Data Center Map n.d.). These regions not only host critical digital infrastructure but are also grappling with chronic water scarcity, amplifying tensions between industrial demand and public resource availability. Public scrutiny has grown in recent years, as seen in controversies surrounding Meta's data center water usage (Southern Environmental Law Center 2025; New York Times 2025).

"The drinking water used in data centers is often treated with chemicals to prevent corrosion and bacterial growth, rendering it unsuitable for human consumption or agricultural use. This means that not only are data centers consuming large quantities of drinking water, but they are also effectively removing it from the local water cycle" (University of Tulsa 2025).

Water pricing structures further complicate this issue. Since water rates are often determined by public authorities based on factors like infrastructure maintenance and treatment costs, "tech companies, such as those operating data centers, pay the same amount for water regardless of their consumption levels" (University of Tulsa 2025). Consequently, these companies can sometimes secure advantageous rates or benefit from pricing systems that fail to account for the true marginal costs of their water consumption. This reduces the financial incentive for data center operators to implement water-saving technologies or more sustainable cooling systems, as they do not bear the full economic burden of their water use (University of Tulsa 2024).

Visualization 1 — Data Center Prevalence by State



Methodology

The data workflow involves cleaning and aggregating water stress metrics, categorizing states by data center density, and calculating composite risk indices using Rstudio and related packages. The visualizations are designed to reveal spatial patterns, proportional stress distributions, and identify high-risk states where AI infrastructure intersects with water scarcity, and demand, i.e. overall water stress.

Why Multiply Data Centers × Mean BWS? The composite risk index amplifies the intersectionality of infrastructure and environmental stress, positioning Arizona, California, and Texas as priority concern zones for sustainable AI deployment.

- A state with many data centers but low water stress = Low Composite Risk.
- A state with few data centers but high water stress = Also Low Composite Risk.
- A state with many data centers in high water stress zones = High Composite Risk.

Example Code:

Mean and Mode BWS Score to show overall water stress by state.

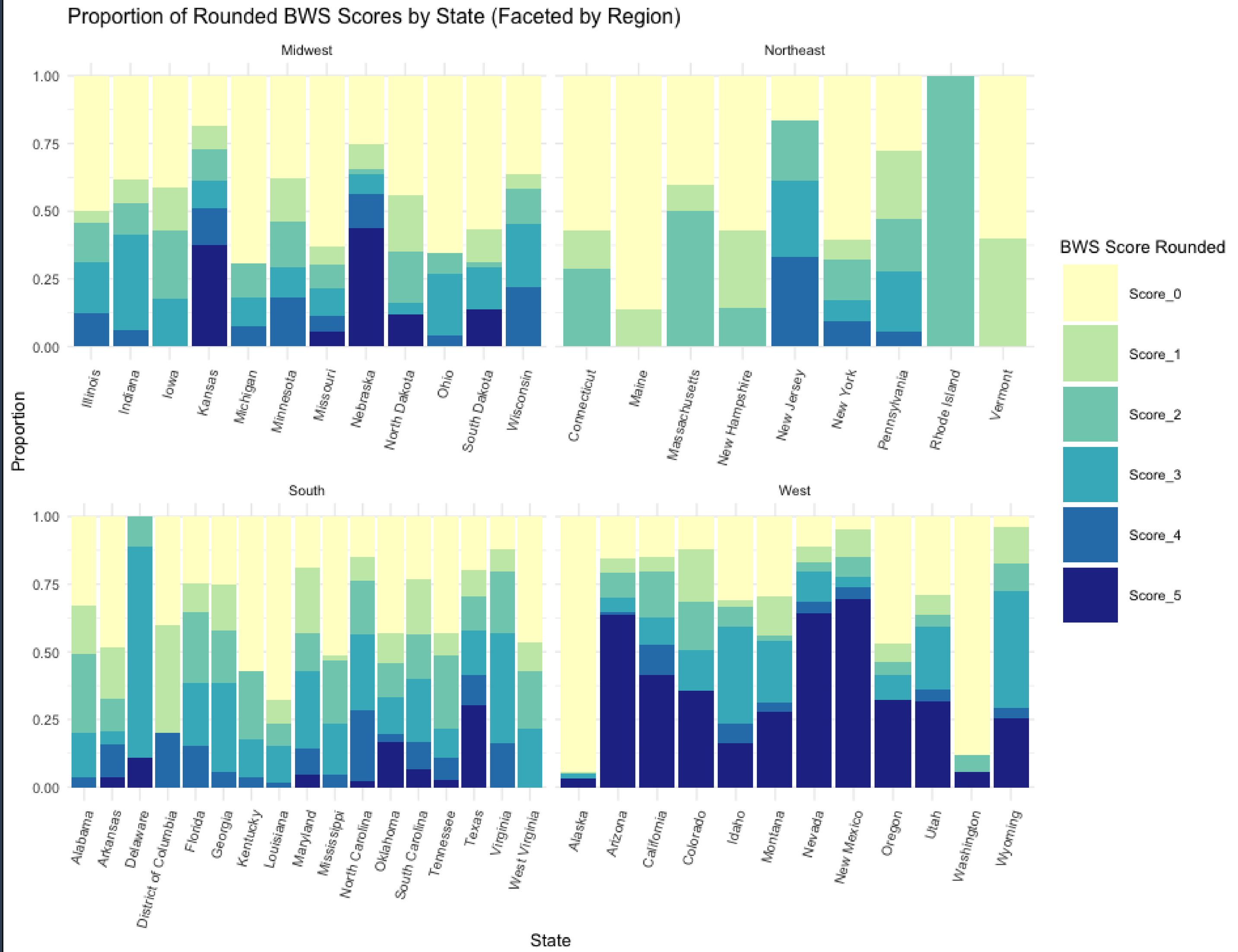
```r
{r}
# Compute Mean BWS Score per State
bws_mean <- WSI %>%
  group_by(name_1) %>%
  summarise(mean_bws = mean(bws_score, na.rm = TRUE))

# Compute Mode BWS Score per State
bws_mode <- WSI %>%
  group_by(name_1, bws_score) %>%
  summarise(count = n(), .groups = 'drop') %>%
  group_by(name_1) %>%
  slice_max(count, n = 1, with_ties = FALSE) %>%
  select(name_1, mode_bws = bws_score)

# Merge Mean and Mode together
bws_summary <- left_join(bws_mean, bws_mode, by = "name_1")

#rename 'name_1' col to 'State' to match data_centers_clean
bws_summary <- bws_summary %>%
  rename(State = name_1)
```

Visualization 2 — Proportional Water Stress Scores by State, then Faceted by Region



Results

Visualization 1 — Data Center Prevalence by State
The heatmap bar chart reveals a steep concentration of data centers in Virginia (600 centers), Texas (378), and California (318). This indicates a significant centralization of AI infrastructure in a few key states, amplifying localized resource demands.

Visualization 2 — Proportional Water Stress Scores by State, then Faceted by Region
By mapping the proportion of areas within each state that fall into rounded Baseline Water Stress (BWS) score categories, we observe stark disparities. Western states such as Arizona and Nevada display a high proportion of regions with BWS scores ≥4, indicating acute localized water stress. Conversely, states like Washington and Michigan show a predominance of low-stress areas.

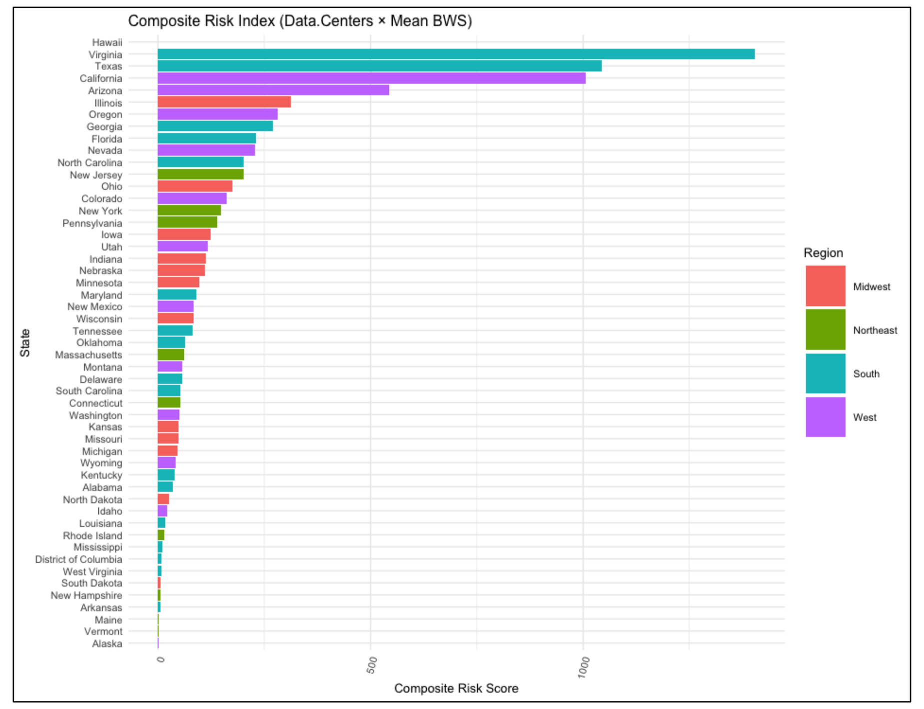
When visualized by region, patterns emerge indicating that Western states are disproportionately represented in higher stress categories. The South, while containing states with extensive data center infrastructure, shows more variability in stress distribution, suggesting both opportunities and vulnerabilities.

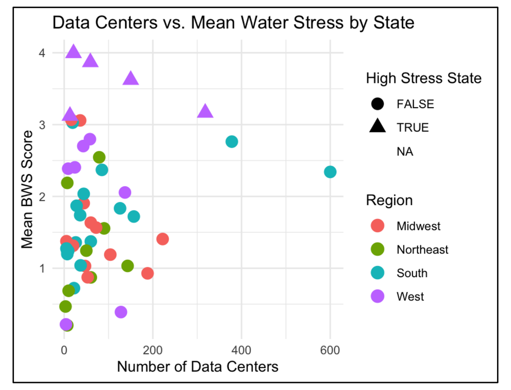
Visualization 3 — Data Centers vs Mean BWS Score
Plotting the number of data centers against the mean BWS score per state highlights several risk profiles. Virginia, despite its large number of data centers, maintains a moderate BWS score, whereas Arizona and Nevada present a more concerning convergence of infrastructure density and high water stress. States like Oregon and Washington, with lower BWS scores, emerge as potentially more sustainable alternatives for future data center expansion.

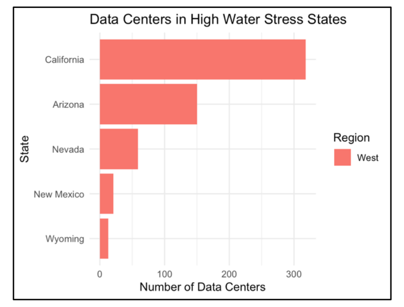
Visualization 4 — Data Centers in High Water Stress States
Filtering for states where the mean BWS score exceeds 3.1 reveals infrastructure concentrations in environmentally vulnerable zones. This visualization underscores the exposure of states like Arizona, California, and Texas to compounded environmental and infrastructure risks.

Visualization 5 — Composite Risk Index (Data Centers × Mean BWS)
The Composite Risk Index amplifies these insights by quantifying the intersection of infrastructure scale and environmental stress. Arizona emerges as a critical concern, closely followed by California and Texas, suggesting that these states represent the highest compounded risk zones for AI infrastructure amid water scarcity challenges.

Visualization 6 — Choropleth Overlay Map
The choropleth map visualizes state-level water stress gradients, overlaid with data center prevalence represented as proportional bubbles. This spatial visualization makes evident the clustering of AI infrastructure in environmentally stressed zones, while also highlighting regions like the Pacific Northwest that could serve as lower-risk alternatives.







Conclusion

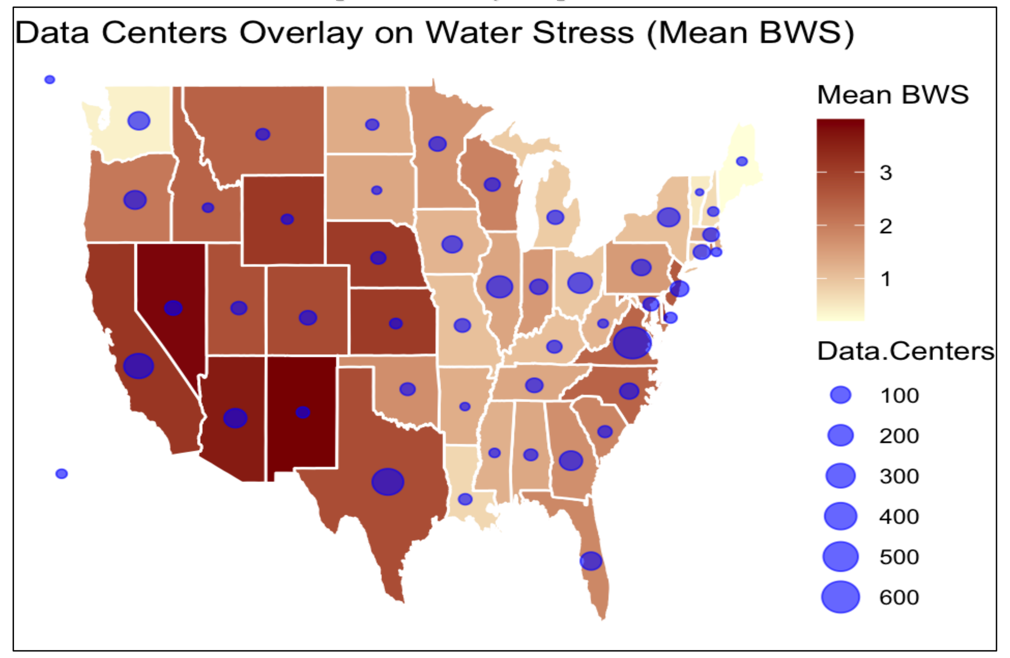
This analysis demonstrates a significant convergence of AI data center infrastructure with regions facing high water stress, particularly in Southern and Western U.S. states. By quantifying and visualizing these intersections, the study provides a data-driven framework to prioritize areas of concern for sustainable infrastructure planning.

The findings suggest that while some high-density data center states like Virginia maintain moderate water stress, others such as Arizona and California are situated in highly vulnerable environmental contexts. These insights are critical for policymakers, utility planners, and corporate ESG strategists aiming to balance infrastructure growth with environmental stewardship.
States with lower water stress but sufficient capacity (e.g., Washington, Oregon) present opportunities for more sustainable AI infrastructure development. However, future expansion strategies must incorporate dynamic water stress projections and enforce transparent reporting of resource usage by tech companies.

Future Improvements
To refine this analysis, future work could incorporate:

- **Data Center Categorization**, distinguishing between data centers based on AI workload intensity (e.g., training hubs vs inference nodes), to better assess resource demands.
- **Weighted Water Stress Indices**, accounting for population density or land area, to emphasize human-relevant impacts.
- **Temporal Projections**, using future water stress scenarios (2030, 2050) to anticipate long-term infrastructure risks.
- **Granular Analysis at the Watershed or County Level**, capturing intra-state disparities that state-level averages might obscure.
- **Corporate Water Use Disclosures**, if accessible, would significantly enhance the precision of this analysis.

Visualization 6 — Choropleth Overlay Map



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